

One-year study on the variation of carotenoid antioxidant substances in living human skin: influence of dietary supplementation and stress factors.

Variation in the level of the carotenoid antioxidant substances beta-carotene and lycopene in the human skin of ten healthy volunteers was measured with resonance Raman spectroscopy in an in vivo experiment over the course of 12 months. Information on the lifestyle of the volunteers concerning dietary supplementation and stress factors was obtained daily by the completion of questionnaires. The results showed individual variations in the levels of carotenoid antioxidant substances in the skin of the volunteers, which strongly correlated to specific lifestyles, such as the intake of dietary supplementations rich in carotenoids, and the influence of stress factors. A carotenoid-rich nutrition, based on large amounts of fruit and vegetables, increased the measured carotenoid levels of skin, while stress factors such as fatigue, illness, smoking, and alcohol consumption gave rise to a decrease in carotenoid levels of the skin. These decreases occurred relatively quickly over the course of one day, while the subsequent increases lasted for up to 3 days. During the summer and autumn months, an increase in the level of carotenoids in the skin was measured for all volunteers. The average "seasonal increase" of the carotenoid content in the skin was determined to be 1.26-fold.